Geometry of Linear Programs

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1. If a LP has an optimal solution, then at least one solution can always be found at: **1 / 1 point**

- ○ Any point within the feasible region.
- ○ At the center of the feasible region.
- ◉ the vertices of the feasible region.
- ○ the intersection of lines representing inequalities with the axes.

Correct
Correct: Optimal solution exists at corner points of feasible region.

2. Consider the following objective function **1 / 1 point**

$$\min 4x + 2y + 3z$$

The feasible region is a convex polyhedron with the following vertices A(2, 3, 1), B(4, 1, 2), C(5, 4, 1), D(3, 5, 3), and E(1, 2, 1). we have provided a name for each vertex and its coordinates (x, y, z).

The optimal value for this LP is:

- ○ 17
- ◉ 9
- ○ 34
- ○ 15

Correct
Correct: Optimal value exists at corner points. E(1,2,1) gives the least value for the objective function.

3. Select all that apply for a convex polyhedron in the context of linear programming. **1 / 1 point**

- [x] For any 2 points within the convex polyhedron , the line segment joining the 2 points also lies completely within the polyhedron.

  Correct
  Correct. This is what makes it a convex polyhedron.

- [x] The optimal solution always lies at one of the vertices of the convex polyhedron.

  Correct
  Correct: The polyhedron is nothing but the feasible region of LP , with optimal solution existing at the boundaries of that region.

- [x] Every point within the convex polyhedron , forms a feasible solution.

  Correct
  Correct. The polyhedron is formed by the intersection of all the constraints and thus forms the feasible region for a given LP.

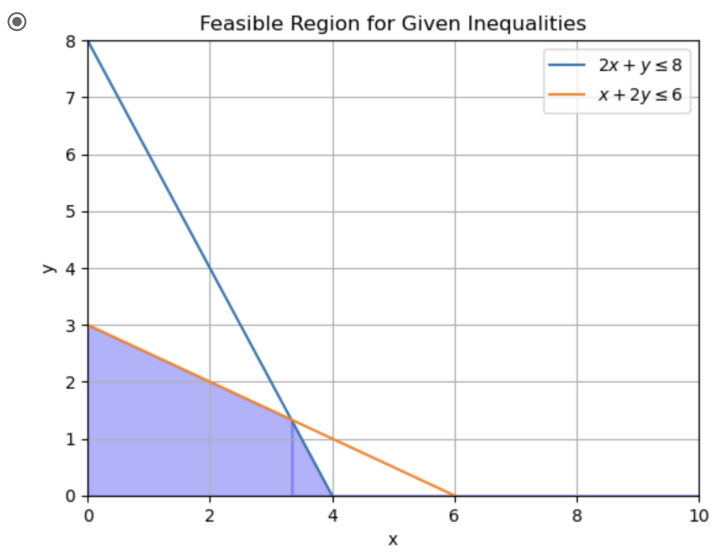
- [x] Convex polyhedron is formed by the intersection of half spaces defined by constraints.

  Correct
  Correct: Constraints form half spaces(divides the space in 2 halves) , and intersection of these half spaces forms the convex polyhedron.

- [ ] The feasible region could be non-convex.

4. Which of the following graphs correctly depicts the feasible region for the given LP : **1 / 1 point**

$$\begin{aligned} \max \quad & 4x + 3 \\ & 2x + y \le 8 \\ & x + 2y \le 6 \\ & x \ge 0 \\ & y \ge 0 \end{aligned}$$

◉


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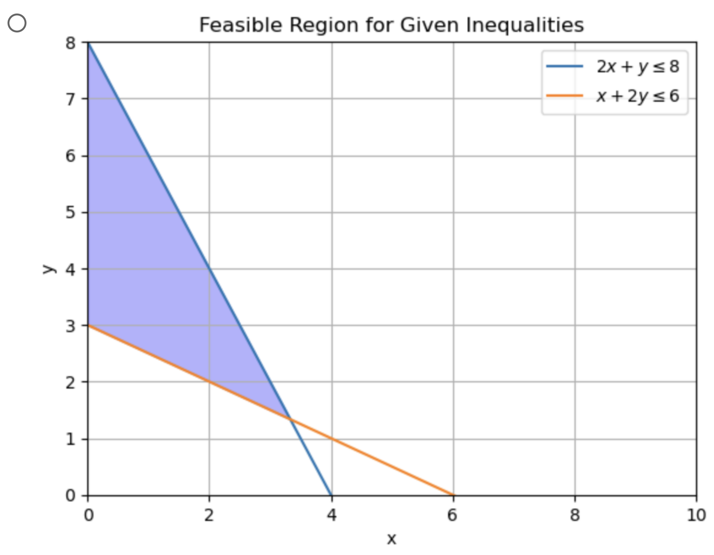

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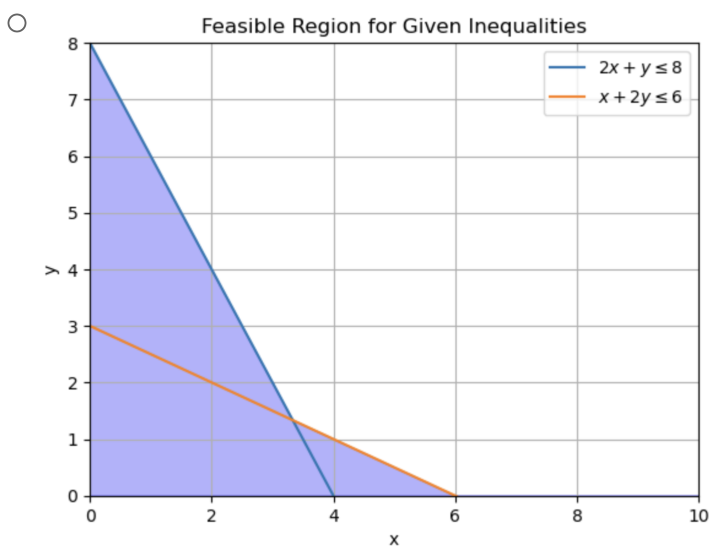

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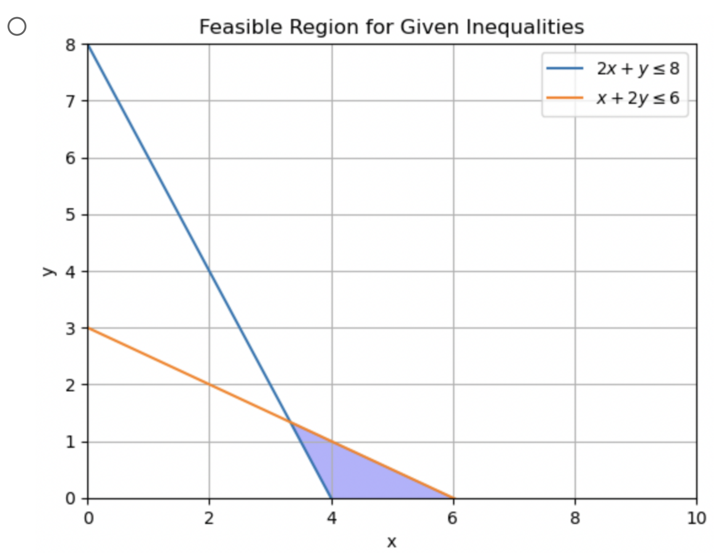

Correct